⚡ Ignition: Reading

Small Engines Overhaul · Unit 6 · Level 3 (closer to the class work) · goes with the Ignition Handout and magneto ignition topic

Name: _____

Ignition: Making the Spark

The power stroke begins with a spark. In most small engines, the spark comes from a magneto, a system that makes its own electricity, so no battery is needed. Permanent magnets are cast into the flywheel. Each time they pass the armature, a coil mounted beside the flywheel, they create a current.

The armature has two windings. The primary winding makes a small current; the secondary winding, with many more turns of wire, boosts it to thousands of volts. This high voltage travels through the plug wire to the spark plug and jumps the gap between the center and side electrodes, igniting the compressed mixture.

Two gaps must be set to the manual's spec: the spark plug gap, checked with a gap gauge, and the armature air gap between the coil and flywheel, set with a gauge card. The flywheel key, a soft metal key, positions the flywheel for correct timing. If a mower blade hits a rock, the key can shear, causing hard starting or kickback.

The kill switch stops the engine by grounding the coil. To check the ignition, disconnect the plug wire and use a spark tester; a bright blue spark is good. Reading the spark plug's color also helps diagnose the engine.

Check your reading

1. Why doesn't a small engine need a battery for spark?

2. What two gaps must be set?

3. What does a sheared flywheel key cause?